

Lessons Learned from PROEMS

From Research Survey to Routine Registry Activity

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theAlfred



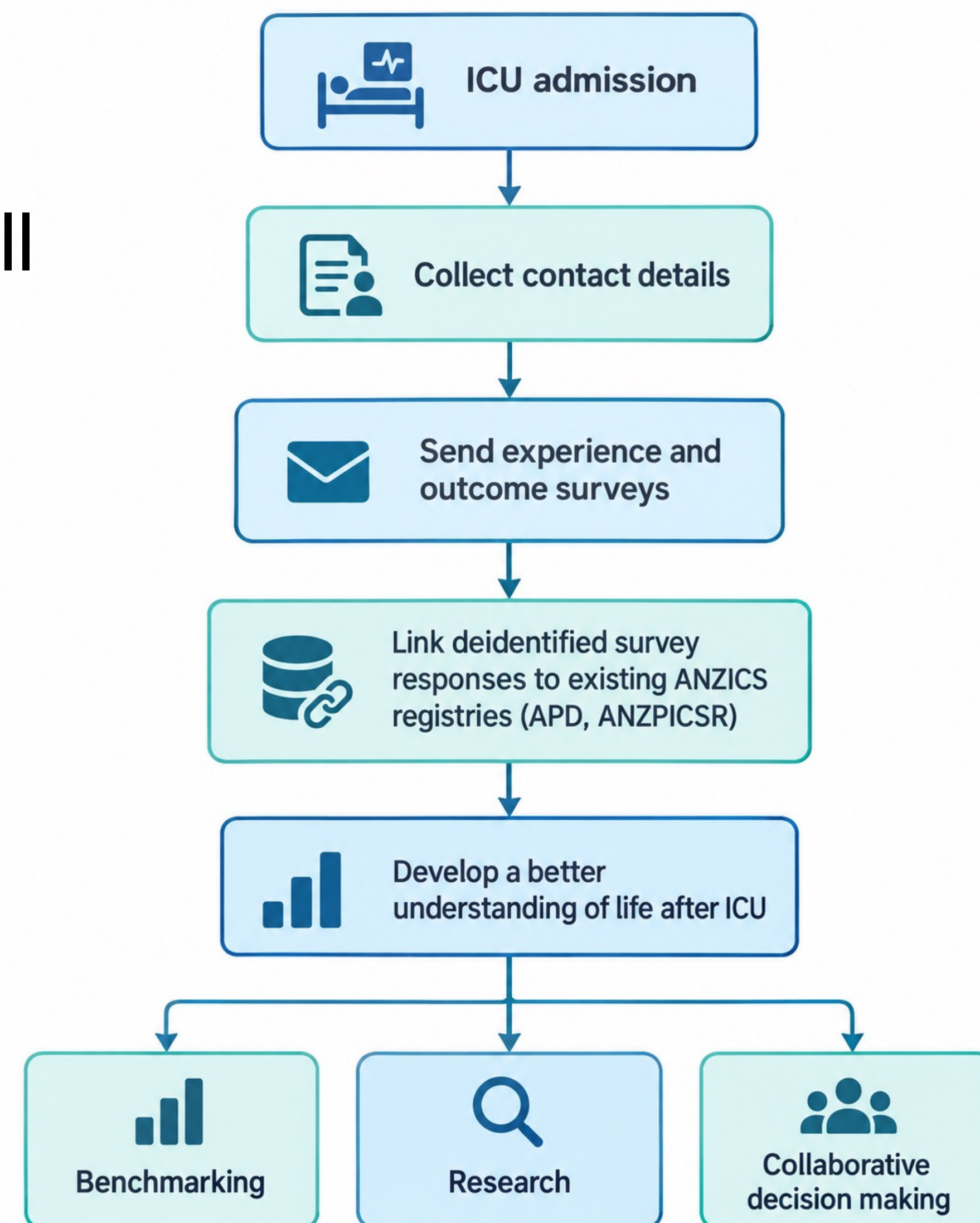
Australian Government

The Idea

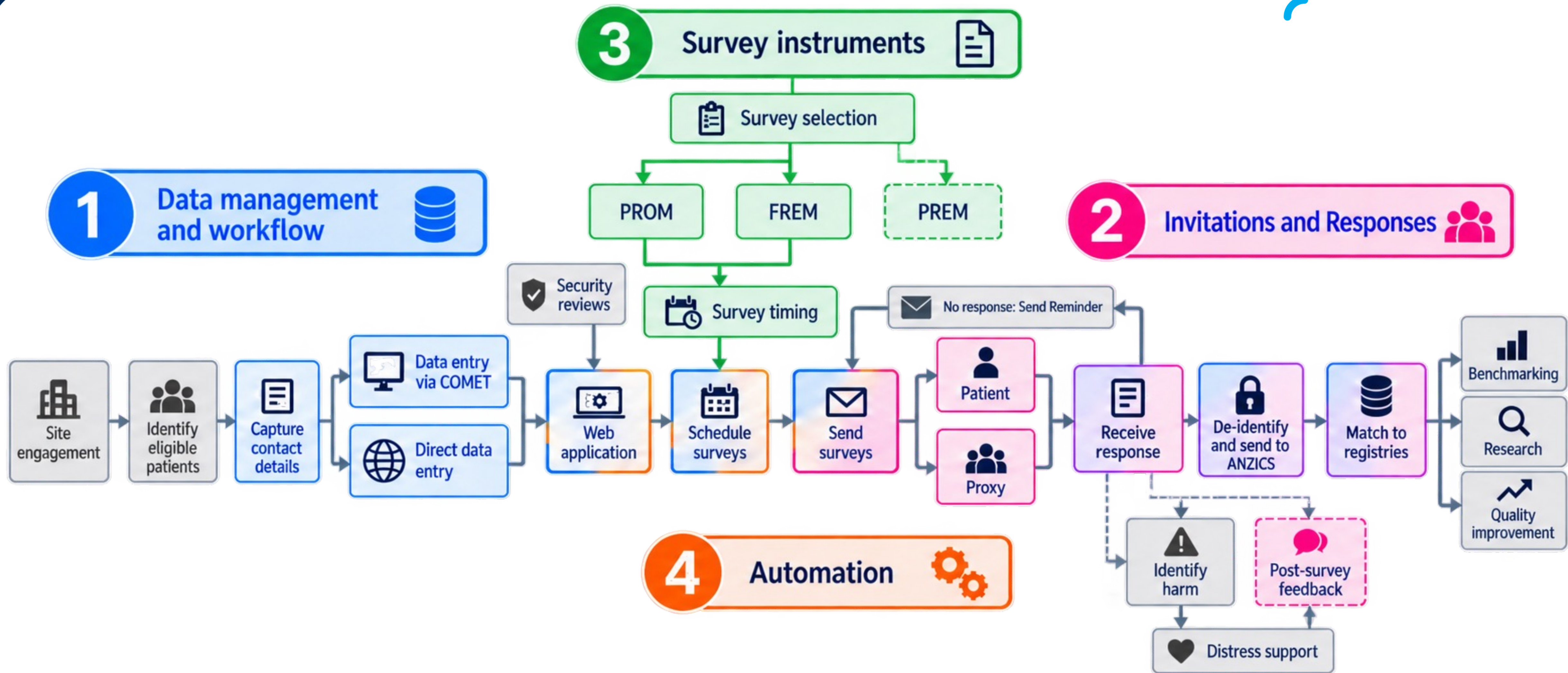


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- We don't know enough about life after ICU
- Most studies on PROMs and PREMs are small
- Most large studies are targeted
- Text messages are cheap
- Registries do ongoing data collection at scale



The Implementation



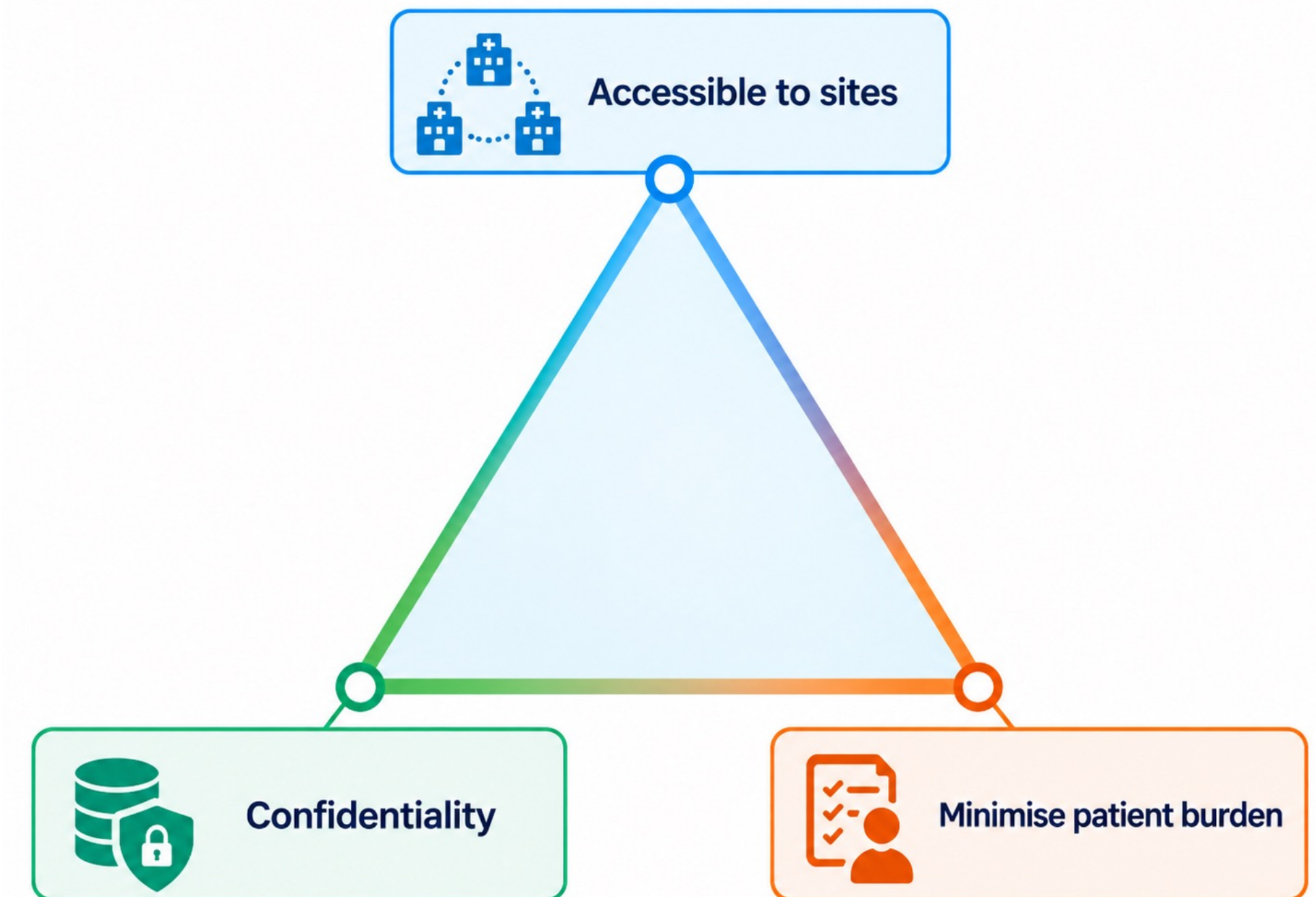
Data Management and Workflow



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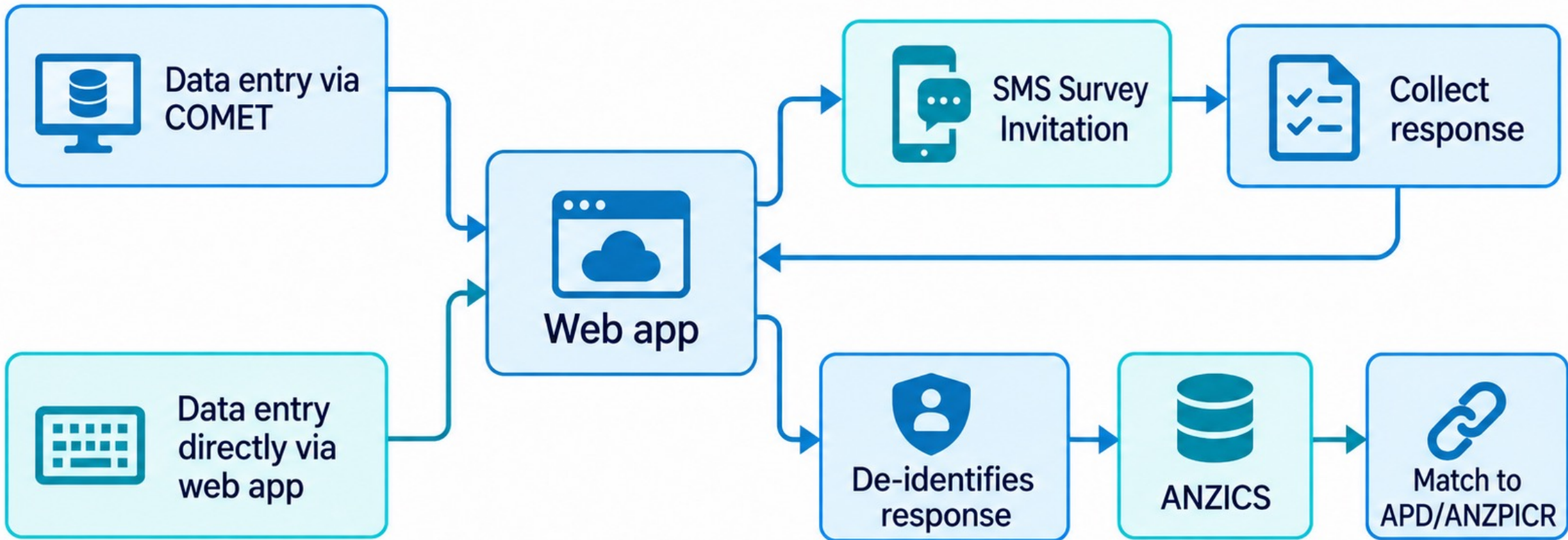
Considerations:

- Huge variation in site data workflows
- Contact details are sensitive health information
- Minimise patient burden



Lessons:

- Reliant on linkage to APD/ANZPICR for meaningful responses
- Increases complexity of system
- High match rates for all instruments
- Lag time for APD data



		PROMs	FREMs-Alive	FREMs-Died
All Records	Responses, n	28,173	21,497	1,460
	Matched, n (%)	27,788 (98.6)	20,866 (97.1)	1,430 (97.9)
Complete	Responses, n	10,306	7,237	584
	Matched, n (%)	10,197 (98.9)	7,147 (98.8)	575 (98.5)

Invitations and Responses



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Considerations:

- How many admissions get invited?
- How many patients will respond?
- Who are the respondents?

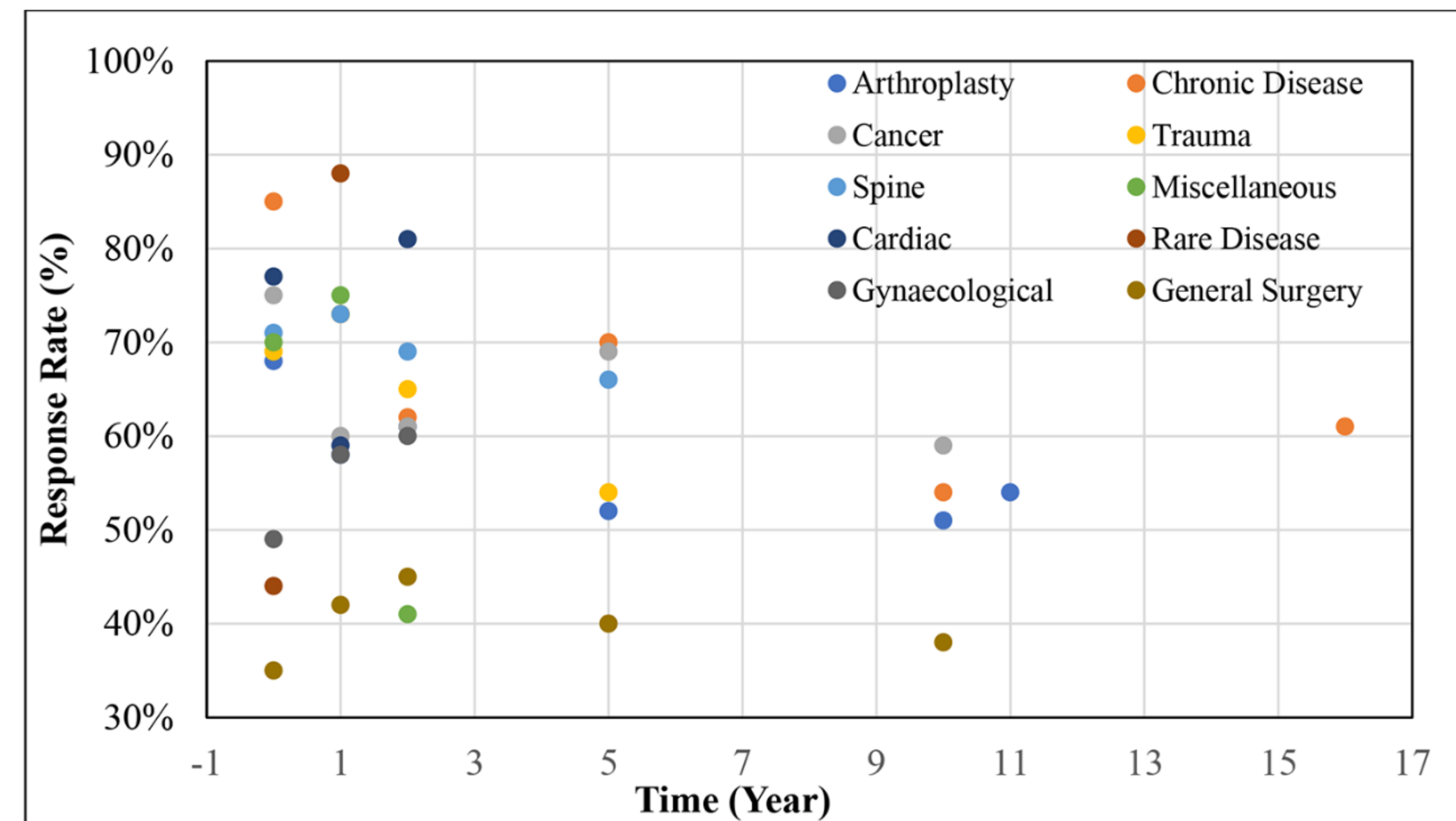


Fig. 3 Average PROMs response rates over time according to health condition

Ruseckaite R, Mudunna C, Caruso M, Ahern S. Response rates in clinical quality registries and databases that collect patient reported outcome measures: a scoping review. Health Qual Life Outcomes. 2023 Jul 11;21:71. doi:[10.1186/s12955-023-02155-5](https://doi.org/10.1186/s12955-023-02155-5) PubMed PMID: 37434146; PubMed Central PMCID: PMC10337187.

Lessons:

- Attrition at both invitation and response stages
 - One-third of admissions not invited
 - One third of those invited respond

Survey attrition by instrument and site COMET status						
	PROM		FREM Alive		FREM Died	
	COMET	Non-COMET	COMET	Non-COMET	COMET	Non-COMET
Eligible admissions	9,413	5,523	10,591	4,385	553	351
Invitations	7,199 (76.5%)	2,682 (48.6%)	5,883 (55.5%)	2,991 (68.2%)	439 (79.4%)	227 (64.7%)
Responses	2,485 (34.5%)	734 (27.4%)	1,907 (32.4%)	795 (26.6%)	167 (38.0%)	58 (25.6%)

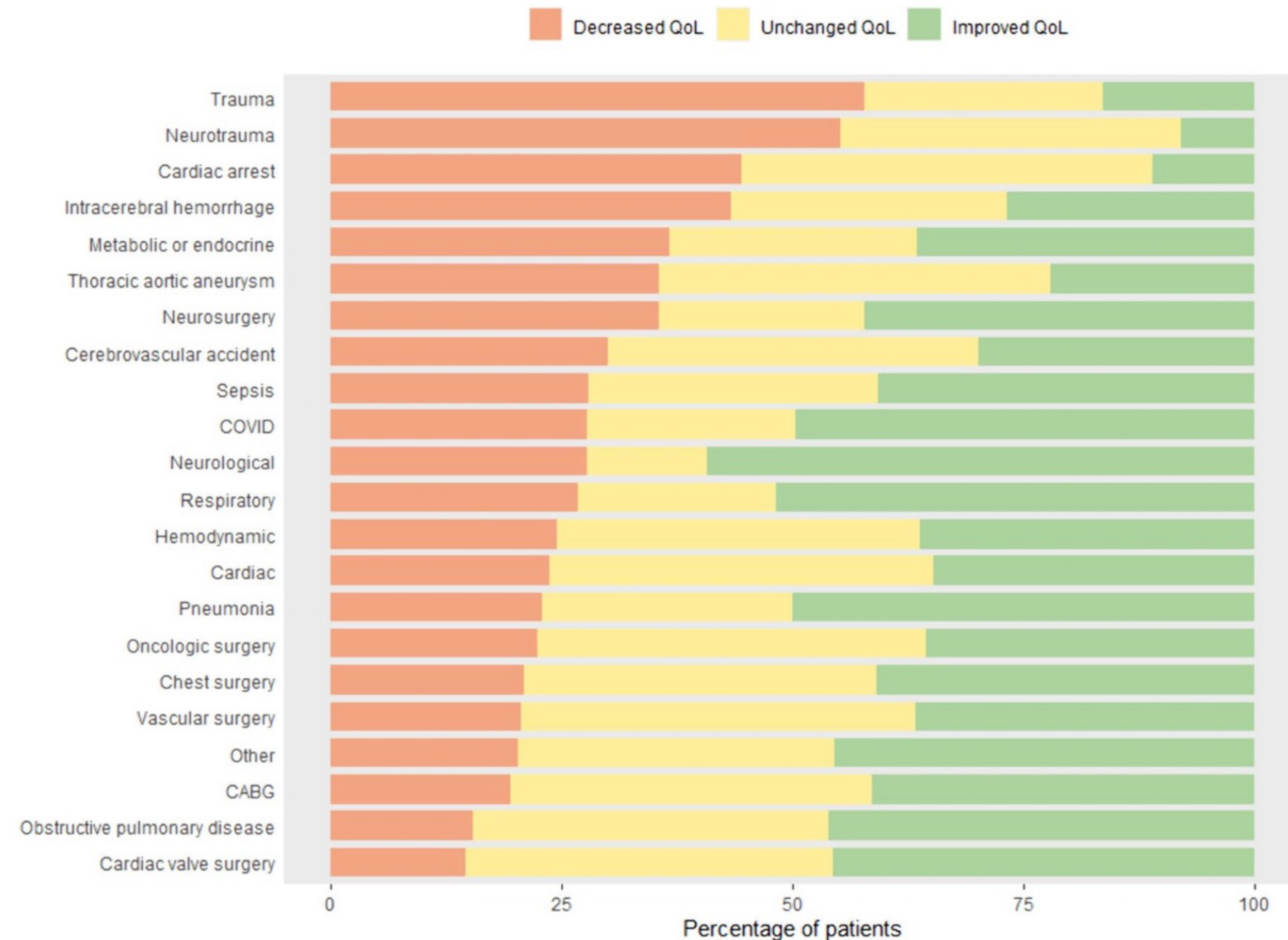
Survey Instruments



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Considerations:

- When do we ask?
- How many times?
- What questions? Free text?
- Who – patients, proxies, both?



Lessons:

- Formal declining is rare
- Survey length not a big issue
- Responses more common from well patients
- Sicker patients had more proxy responses
- Sites and recipients *really* want free-text

Survey response status	
Complete	6,039 (31.1%)
Partially complete	107 (0.6%)
Incomplete	7 (0.0%)
Declined consent	362 (1.9%)
Withdrew consent	19 (0.1%)
No response	12,887 (66.4%)
Immediate response	3,563 (58.0%)
PROM, FREM Alive, and FREM Died survey records are pooled. Response status rows are n (%) of all survey records.	

Patient characteristics by PROM respondent type		
Characteristic	Patient N = 2,341	Proxy N = 836
Age (years)	65.3 (53.7, 73.5)	66.1 (51.7, 75.8)
Female	1,052 (45%)	331 (40%)
Indigenous	46 (2.0%)	33 (3.9%)
ANZROD risk of death (%)	4.0 (9.4)	7.2 (13.7)
ICU length of stay (days)	1.7 (0.9, 3.0)	2.0 (1.0, 3.9)

Benefits and Pitfalls of Automation



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Considerations:

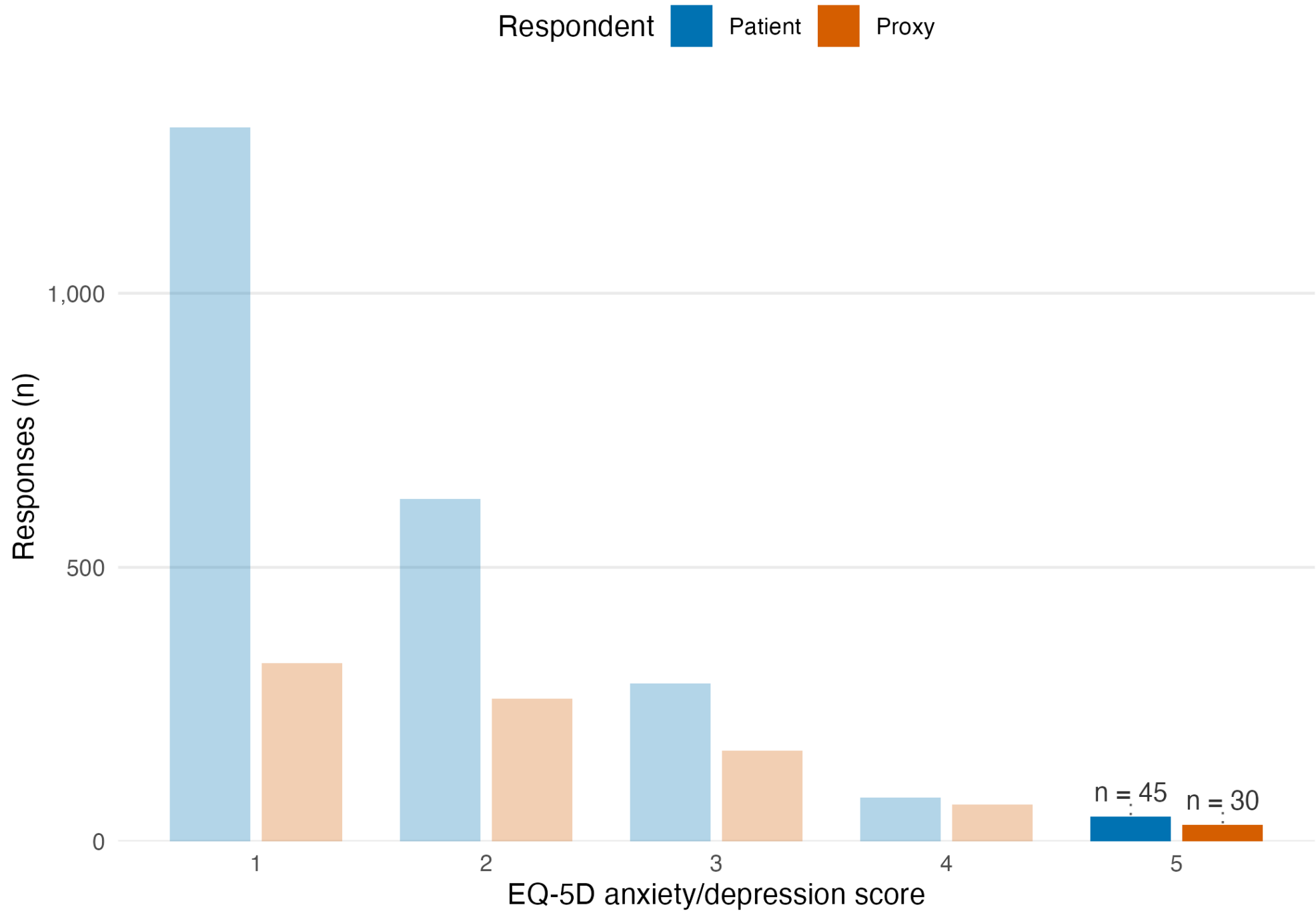
- Automated electronic messaging makes this possible
- What are the edge cases?
- Will the collection process be harmful?
- How do we manage follow-up?

Benefits and Pitfalls of Automation

Lessons:

- Benefits grow with scale
 - Errors do too
- System should be robust to errors
- Consider a “Break glass” mechanism

Economic Outcomes	Amount
Capital Expenditure	
Application Development	\$430,705.00
ANZICS Operational Expenditure	
Information Technology Costs	\$2,000.00 per month
Project Staff	\$7,608.00 per month
Administration support	\$4,262.00 per month
Text Message Costs	\$15.00 per site per month





Future work:

- Characterise selection and response bias?
- How can we improve invitation and response rater?
- Can we survey patient experience shortly after ICU discharge?
- How can we manage free-text responses?
- Break-glass mechanism for site review



Thank you

- Wilson Low
- Shaila Chavan
- Sue Huckson
- Jennifer Hogan
- David Pilcher
- Edward Litton
- Paul Secombe
- Canberra Hospital
- Fairfield Hospital
- Liverpool Hospital
- University Hospital Geelong
- Sunshine Hospital
- Gold Coast University Hospital
- Alfred Hospital
- St John Of God Health Care (Subiaco)
- Campbelltown Hospital
- North Shore Private Hospital
- Nepean Hospital
- Dunedin Hospital
- Bendigo Health Care Group
- Fiona Stanley Hospital
- Townsville University Hospital
- Wellington Hospital
- Alice Springs Hospital

The Implementation



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Governance

Ethics (Australia and New Zealand)

Site engagement

Site representativeness

Security reviews

Consent/opt-out

Distress and bereavement support

Data management

HPP compliance

Identified data for survey invitation

Deidentified data for ANZICS staff

Linkable to APD for meaningful analysis

**Inviting and sending surveys
COMET vs non-COMET sites**

Exception management

Survey delivery

Number of surveys

Survey timing

Survey length

Patient awareness

Automated delivery

Response rates

Proxy responses

Language